THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS · HARD LEVEL UNIT TEST SET 1

Chapters 1–5: **Patterns in Mathematics, Lines and Angles, Number Play, Data Handling and Presentation, Prime Time** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. In a party 15 people greet each other once with a hug. Number of hugs: 1

(a) 105 **(b)** 90 **(c)** 106 **(d)** 120

2. At 1:10 the smaller angle between the clock hands is: 1

(a) 25 **(b)** 40 **(c)** 30 **(d)** 20

3. How many steps does the Kaprekar routine take to reach 6174 from 3141? 1

(a) 4 **(b)** 6 **(c)** 7 **(d)** 5

4. The total of the table, after removing the largest value, is: 1

Item	Number
Orange	25
Grapes	45
Guava	30
Banana	25
Apple	80
Mango	55

(a) 175 **(b)** 180 **(c)** 190 **(d)** 185

5. The smallest 4-digit number divisible by 13 is: 1

(a) 1001 **(b)** 1002 **(c)** 1014 **(d)** 988

6. The sum $1 + 2 + 3 + \cdots + n$ equals 703. Then n is: 1

(a) 39 **(b)** 36 **(c)** 38 **(d)** 37

7. A wheel turns 3 full turns and then a further quarter turn. The total angle in degrees is: 1

(a) 1080 **(b)** 1170 **(c)** 1260 **(d)** 1350

8. The Collatz sequence of 9 has how many terms (including 9 and 1)? 1

(a) 21 **(b)** 22 **(c)** 20 **(d)** 19

9. One symbol of a pictograph stands for 10. Class A has 12 symbols and its total is 3 times Class B. How many symbols does Class B have? 1

(a) 3 **(b)** 6 **(c)** 4 **(d)** 5

10. The number of composite numbers between 1 and 30 is: 1

(a) 19 **(b)** 20 **(c)** 18 **(d)** 21

11. The sum of the first 30 odd numbers is 900. The sum of the odd numbers from 21 to 59 is: 1

(a) 810 **(b)** 800 **(c)** 799 **(d)** 801

12. A spinner starts pointing North and turns clockwise 270° then anticlockwise 90°. The final angle clockwise from North is: 1

(a) 270 **(b)** 190 **(c)** 170 **(d)** 180

13. The number of 4-digit palindromes with first digit 2 is: 1

(a) 10 **(b)** 9 **(c)** 11 **(d)** 19

14. A bar graph has scale 1 cm = 12 units. The heights of two bars differ by 7 cm. The values differ by: 1

(a) 96 **(b)** 72 **(c)** 84 **(d)** 89

15. The smallest number divisible by each of 6, 8 and 12 is: 1

(a) 30 **(b)** 25 **(c)** 24 **(d)** 23

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. 14 cricket teams play a league in which every team plays every other team once. Find the number of matches. 2

17. At what angle are the clock hands at 4:10? 2

18. How many steps does the Kaprekar routine need for 1234? Write the chain. 2

19. In a survey of 150 people, 62 like tea, 48 like coffee, and the rest like milk. Find the number liking milk and show data in a table. 2

20. Find the smallest number divisible by 8, 12 and 18. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. In a pattern the 1st term is 14, the 2nd is 23, and so on by adding 9. Which term of the pattern equals 365? Find also the term just before it. 3

22. At 1:50 how far (in degrees) is the minute hand ahead of the hour hand, and how far is it from 12? 3

23. Find a 4-digit palindrome that is divisible by 11 and has digit sum 12. Verify. 3

24. The table shows two weeks of sales (units). Find the week with the larger total and by how much. 3

Day	Week 1	Week 2
Mon	13	15
Tue	20	19
Wed	11	14
Thu	17	12

OR

Three traffic lights change every 48, 72 and 108 seconds. If they change together at 8:00 am, after how many seconds do they next change together and at what time? 3

25. Sums of consecutive odd numbers give squares. Find the sum of the odd numbers from 1 to 39 and use it to find 20^2. Then evaluate the sum from 41 to 49. 3

OR

Find the time after 2:00 when the minute hand has turned 120° and then state the smaller angle between hands. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. Look at the pattern: the 8th figure has 46 dots. How many dots are in figure 15? If a figure has 118 dots, which figure is it? Show that the dots form a pattern with common difference 6. 5

OR

At 1:10, find 5
(a) the angle turned by the hour hand since 12 o'clock,
(b) the angle turned by the minute hand since 12,
(c) the smaller angle between the hands,
(d) its type.

27. A palindromic number of 5 digits is formed as $abcba$ where $a = 1$, $b = 2$, $c = 3$. Write the number, find its digit sum, check if it is divisible by 9, and find the smallest number to add to make it divisible by 9. 5

OR

A bar graph has scale 1 cm = 12 units. Bars A, B, C have heights 3 cm, 6 cm, 9 cm. D's value is equal to the average of A, B, C. Find the values and the height of bar D. 5

END OF QUESTION PAPER Total: 50 Marks

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